Assignment 1: Minimal Spanning Trees in Medical Image Processing

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Course: Computational Mathematics for Engineers (Minor)

1 Introduction

1.1 Overview of MST Applications in Image Processing

This report explores the application of Minimal Spanning Trees (MSTs), specifically Kruskal's algorithm, to medical image segmentation and denoising. An image can be viewed as a weighted grid graph where vertices represent pixels and edges represent adjacency relationships. By utilizing these graph representations, MST algorithms can efficiently group similar pixels together.

1.2 Motivation for Medical Image Segmentation

Medical image segmentation is crucial for diagnosis and treatment planning. Given a gray-scale medical image, the objective is to identify distinct anatomical structures while reducing noise. Implementing this Kruskal-inspired algorithm demonstrates the connection between graph theory and computational medicine.

2 Methodology

2.1 Implementation Approach

The algorithm processes a gray-scale image of size $m \times n$, where each pixel has an intensity value $p(i, j) \in [0, 255]$.

A graph is constructed as:

$$G = (V, E)$$

where V represents the pixels and edges connect 4-neighbor pixels.

$$E = \{(v_{i,j}, v_{k,l}) \mid |i - k| + |j - l| = 1\}$$

The edge weight captures dissimilarity between neighboring pixels:

$$w(v_{i,j}, v_{k,l}) = |I(i,j) - I(k,l)|$$

A modified Kruskal's algorithm is applied for segmentation. Each pixel initially forms a separate component. Edges are sorted in non-decreasing order and processed sequentially.

2.2 Mathematical Formulation of the Segmentation Criterion

Two components C_u and C_v are merged if:

$$w(e) \leq \min(Int(C_u) + \tau(C_u),\ Int(C_v) + \tau(C_v))$$

where:

- $Int(C)$ is the maximum weight in the component's MST
- $\tau(C)$ is a threshold function
- k is a scale parameter controlling segment size

The threshold function is defined as:

$$\tau(C) = \frac{k}{|C|}$$

2.3 Segment-based Denoising

Denoising is performed by replacing pixel intensity with the mean intensity of its segment:

$$I_{denoised}(p) = \frac{1}{|C(p)|} \sum_{q \in C(p)} I(q)$$

where $C(p)$ is the segment containing pixel p.

3 Results and Analysis

3.1 Quantitative Metrics

Image	Scale k	Segments	Time (s)	MSE	PSNR (dB)
Normal Brain (Te-no_1.jpg)	500	693	0.72	869.79	18.74
Glioma (Te-gl_1.jpg)	500	1605	2.10	473.66	21.38
Meningioma (Te-aug-me_1.jpg)	500	1269	3.62	200.62	25.11

3.2 Parameter Sensitivity Analysis

The algorithm exhibits high sensitivity to the scale parameter k. The normal brain image processed fastest but yielded higher MSE due to complex tissue structures. The glioma produced the highest number of segments because the modified Kruskal algorithm preserved irregular tumor borders. The algorithm achieved the highest accuracy on the meningioma image with a PSNR of 25.11 dB due to its sharp and high-contrast boundaries.

4 Discussion

4.1 Advantages and Limitations

The MST-based approach efficiently segments images by preserving edges between different regions while applying segment-based denoising. However, relying solely on local intensity differences may cause leakage across boundaries if strong edges are absent.

4.2 Comparison with Other Methods

Unlike clustering methods that ignore spatial relationships, the MST method ensures spatially contiguous segments. It also serves as a lightweight alternative to deep learning methods since it does not require training data.

4.3 Possible Improvements

Future improvements could include:

- Using an 8-neighbor grid instead of 4-neighbor connectivity
- Applying Gaussian blur preprocessing to reduce noise
- Combining MST segmentation with learning-based refinement

5 Conclusion

This implementation successfully applied Kruskal's algorithm for image segmentation. Using adaptive thresholding for merging decisions allowed the algorithm to isolate anatomical structures while reducing noise. The approach demonstrates how graph theory can be effectively applied in computational medicine.

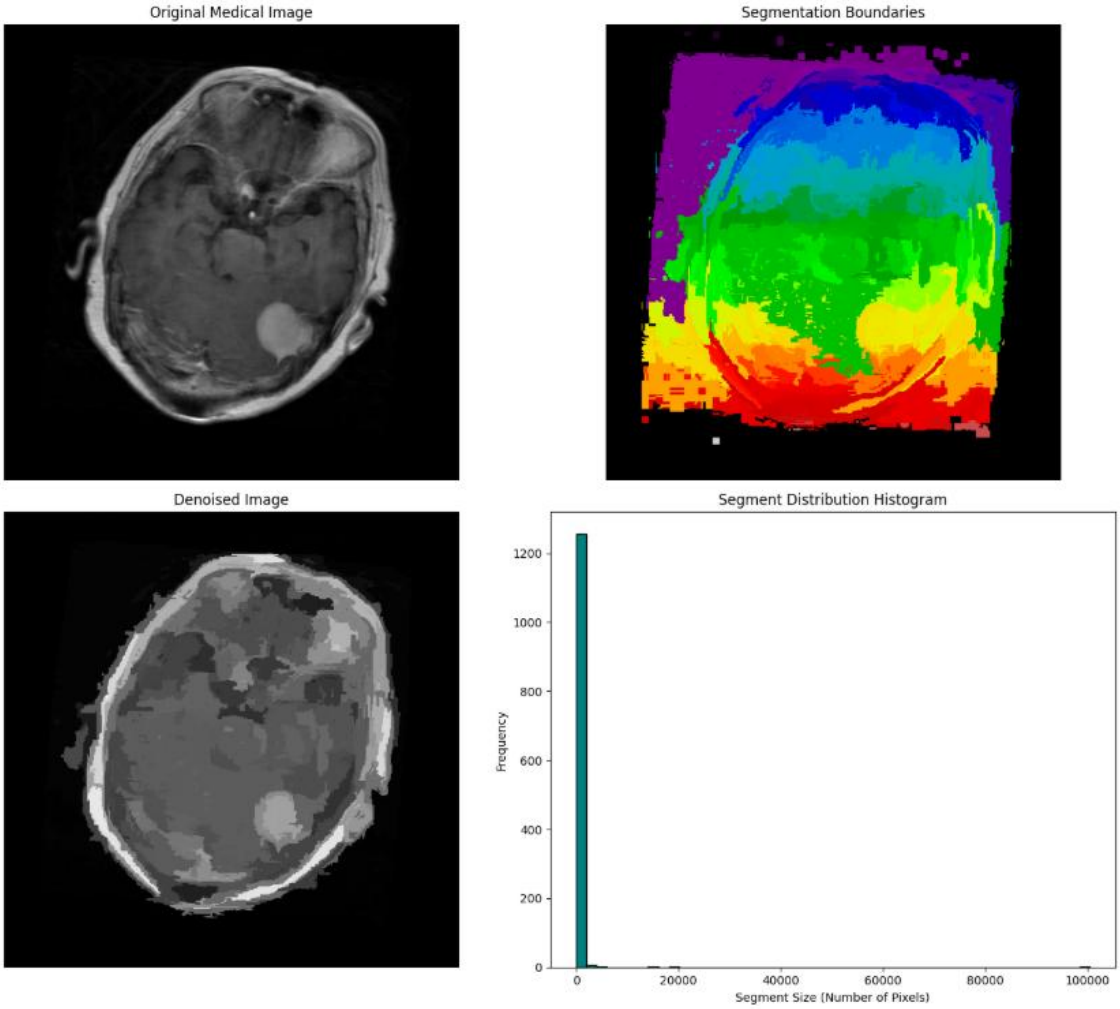
GitHub Repository

https://github.com/amruthaajish17/Assignment_1

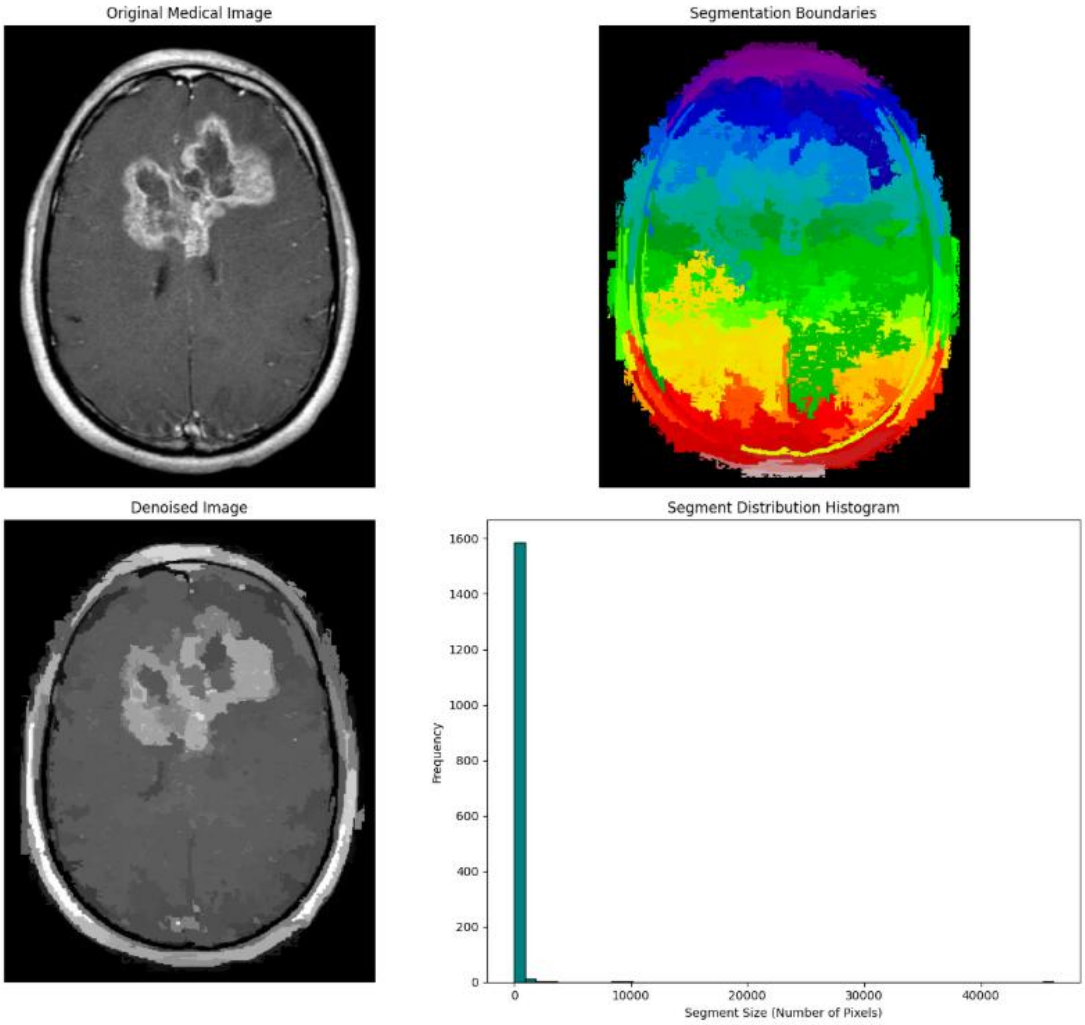
6 Output

OUTPUT:

```
==================================================
PROCESSING IMAGE: /content/Te-aug-me_1.jpg
==================================================
Segmenting image with k=500...
Applying segment-based denoising...
------------------------------
QUANTITATIVE METRICS
------------------------------
Computation Time:   3.43 seconds
Number of Segments: 1269
Mean Squared Error: 200.62
PSNR:               25.11 dB
------------------------------
```



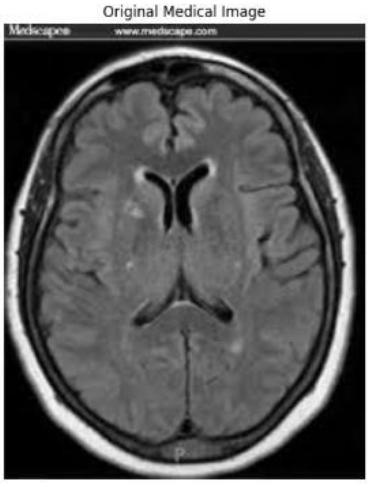
```
==================================================
PROCESSING IMAGE: /content/Te-gl_1.jpg
==================================================
Segmenting image with k=500...
Applying segment-based denoising...
------------------------------
QUANTITATIVE METRICS
------------------------------
Computation Time:   2.22 seconds
Number of Segments: 1605
Mean Squared Error: 473.66
PSNR:               21.38 dB
------------------------------
```



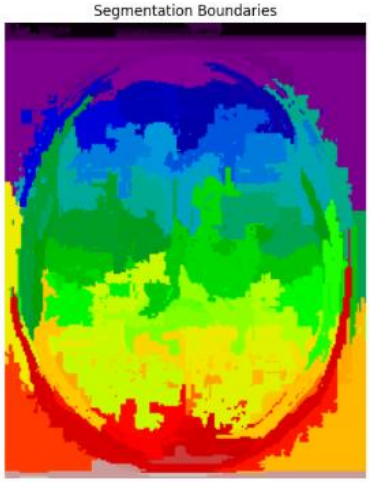
```
Initializing MST-based Image Processing...

==================================================
PROCESSING IMAGE: /content/Te-no_1.jpg
==================================================
Segmenting image with k=500...
Applying segment-based denoising...
------------------------------
QUANTITATIVE METRICS
------------------------------
Computation Time:   0.65 seconds
Number of Segments: 693
Mean Squared Error: 869.79
PSNR:               18.74 dB
------------------------------
```

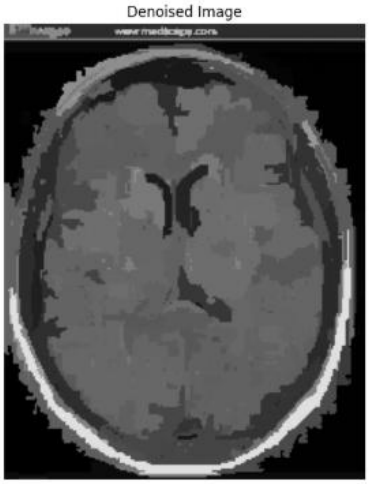
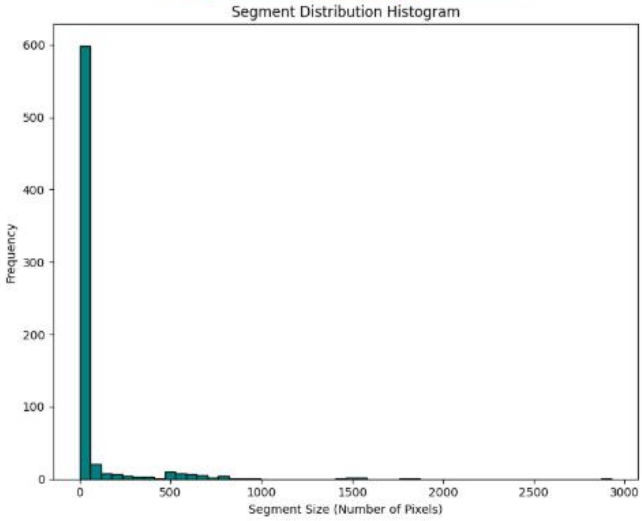
Original Medical Image



Segmentation Boundaries



Denoised Image



Segment Distribution Histogram